THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 4: **Quadratic Equations** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The equation $x^2 + 4x + 4 = 0$ has: 1

(a) two distinct real roots **(b)** no real roots
(c) two equal real roots **(d)** one real root and one imaginary root

2. The larger root of $x^2 - 14x + 45 = 0$ is: 1

(a) 10 **(b)** 9 **(c)** 11 **(d)** 8

3. The sum of the roots of $2x^2 - 7x - 15 = 0$ is: 1

(a) $\frac{9}{2}$ **(b)** $\frac{7}{2}$ **(c)** $\frac{5}{2}$ **(d)** $\frac{11}{2}$

4. The product of the roots of $2x^2 - 7x - 15 = 0$ is: 1

(a) $-\frac{17}{2}$ **(b)** $-\frac{13}{2}$ **(c)** $-\frac{11}{2}$ **(d)** $-\frac{15}{2}$

5. The roots of $x^2 - 15x + 44 = 0$ differ by: 1

(a) 6 **(b)** 7 **(c)** 8 **(d)** 9

6. If $x = 3$ is a root of $x^2 - 10x + k = 0$, then $k =$ 1

(a) 23 **(b)** 20 **(c)** 22 **(d)** 21

7. The equation $x^2 + 3x + 5 = 0$ has: 1

(a) no real roots **(b)** two distinct real roots
(c) two equal real roots **(d)** one real root and one imaginary root

8. The larger root of $x^2 - 16x + 48 = 0$ is: 1

(a) 13 **(b)** 11 **(c)** 12 **(d)** 14

9. The sum of the roots of $3x^2 - 19x + 6 = 0$ is: 1

(a) $\frac{25}{3}$ **(b)** $\frac{19}{3}$ **(c)** $\frac{16}{3}$ **(d)** $\frac{22}{3}$

10. The product of the roots of $3x^2 - 19x + 6 = 0$ is: 1

(a) 3 **(b)** 4 **(c)** 2 **(d)** 1

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. If $x = 5$ is a root of $x^2 - 11x + k = 0$, find k and the other root. 2

12. Find k so that $kx^2 - 8x + 2 = 0$ has equal roots. 2

13. Solve for x: $2x^2 - 7x - 15 = 0$ by the quadratic formula. 2

14. If $x = 3$ is a root of $x^2 - 10x + k = 0$, find k and the other root. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. Solve for x: $\frac{1}{x} + \frac{1}{x+2} = \frac{8}{15}$. 3

16. Find the nature of the roots of $4x^2 - 12x + 9 = 0$ and, if real, find them. 3

17. Solve by completing the square: $2x^2 - 11x + 5 = 0$. 3

OR

Solve for x: $\frac{1}{x} + \frac{1}{x+1} = \frac{9}{20}$. 3

18. Find the nature of the roots of $x^2 - 2x + 5 = 0$ and, if real, find them. 3

OR

Solve by completing the square: $x^2 - 8x - 9 = 0$. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. A train travels 420 km at a uniform speed. If the speed had been 5 km/h more, it would have taken 2 hours less for the journey. Find the original speed. 5

OR

A motorboat goes 24 km upstream and 24 km downstream in 5 hours. If the speed of the stream is 2 km/h, find the speed of the boat in still water. 5

20. A train travels 240 km at a uniform speed. If the speed had been 8 km/h more, it would have taken 1 hour less for the journey. Find the original speed. 5

OR

A motorboat goes 24 km upstream and 24 km downstream in 6 hours. If the speed of the stream is 3 km/h, find the speed of the boat in still water. 5

END OF QUESTION PAPER Total: 40 Marks

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